



The role of microheterogeneity in cell fate decisions in neural progenitors and neural crest

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Neuroprogenitors must integrate a multitude of signals, including gradients of morphogens, transcriptional programs, and temporal cues to generate an astonishing diversity of cell types inhabiting the nervous system. How do these different layers of information come together to influence cell fate in progenitor cells in a coordinated way? Here we provide a nuanced perspective on cell fate selection in the nervous system and neural crest lineage, suggesting that it is not a straightforward, deterministic process governed by rigid on-off switches. Instead, the process involves probabilistic transitions influenced by small variations - termed "microheterogeneity" - within a progenitor cell population. These minuscule differences between individual neural progenitor cells can result in significantly different outcomes, making certain fates more probable for some cells than others. Here we discuss the diversity of such examples and the theory behind, also providing future perspectives.

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Introduction

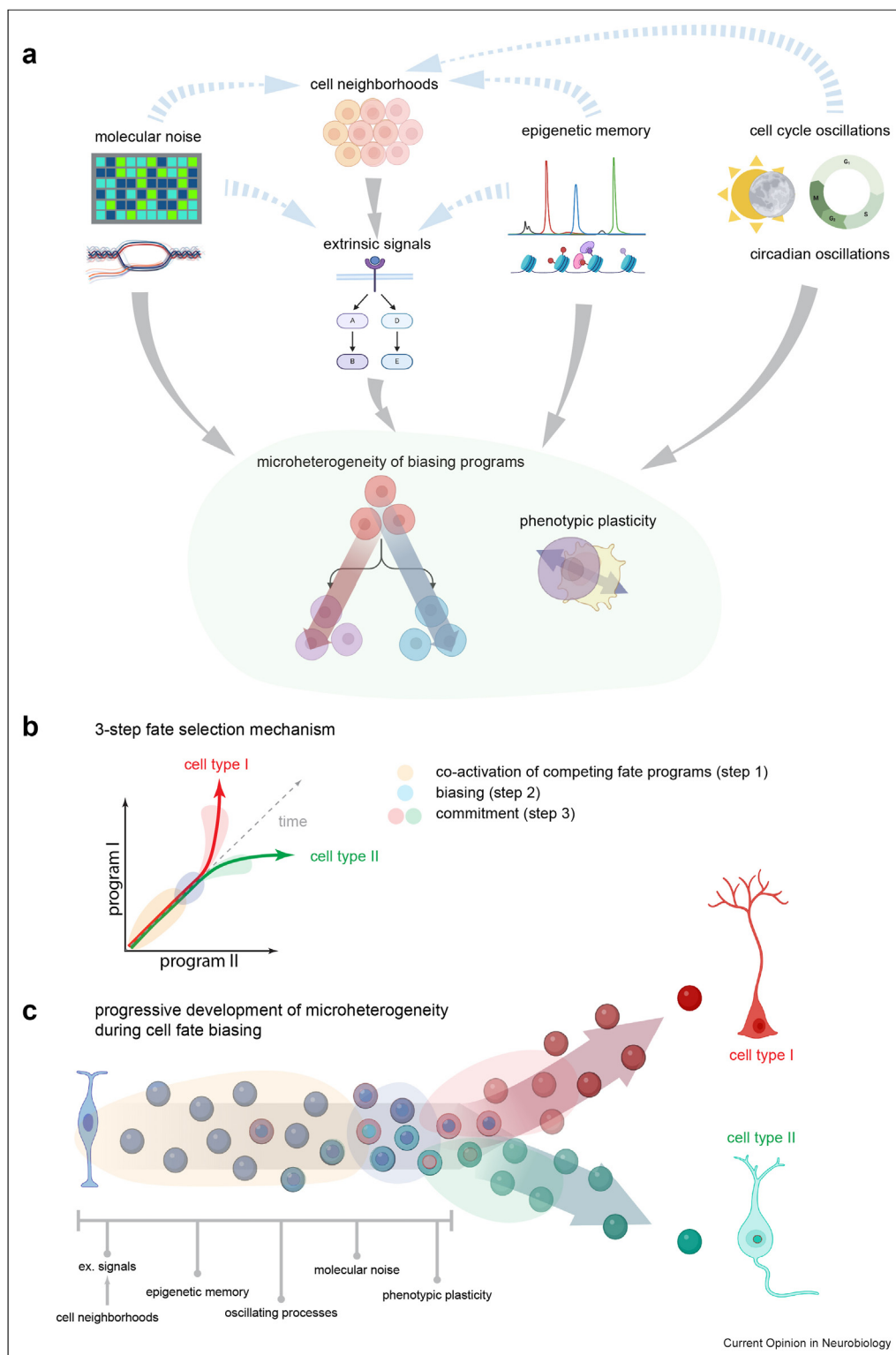
Neural development is one of the most complex and finely tuned processes in biology. The central challenge lies in generating the enormous diversity of neuronal and glial cell types necessary to form the brain, spinal cord, and peripheral nervous system. In this process,

different regions of the nervous system - the neural plate, neural crest, and neurogenic placodes - are patterned early on, and from these regions arise distinct neural progenitors that will give rise to the diversity of cell types [1–3]. But how does a system starting from relatively uniform early progenitors such as neuroectoderm or premigratory neural crest creates such a wide array of specialized cells, ensuring not only the diversity required for function but also balanced proportions, correct positions and proper patterning, preventing over-competition for any particular cell type? We know that neural progenitors receive a complex array of signals that vary by region, time, and context. How do cells interpret these signals in a context-aware manner to generate the appropriate cell types for their location in the nervous system?

One of the earliest conceptual models for understanding cell fate decisions is Waddington's 'landscape' metaphor, introduced in 1957 [4]. This metaphor has been reimagined by modern scientists, depicting cells as 'rolling' from a pluripotent state into a differentiated state, guided by 'valleys' shaped by underlying gene regulatory networks. In contemporary interpretation of this landscape, cells integrate numerous signals and, based on their current state, determine their path of further differentiation. Such signals and conditions include cell neighborhoods and proximity signals, cell migration terrains, biomechanics, morphogens, accumulated cell history and epigenetic states, transcriptional noise and other random fluctuations, as well as cell cycle and circadian oscillations [5,6]. These factors jointly and often simultaneously act on the progenitor cells at different time points driving the development of the spectrum of the terminal cell types [7].

In one of the key examples, the migrating neural crest progenitors generate an astonishing microheterogeneity of fate-biased states at the population level. This microheterogeneity is linked to genetic programs forming a particular fate bias, which is detectable early, even during the process of neural crest cell delamination from the neural tube [8]. The molecular nature of these biases is diverse, encompassing transcription factors, and various other components that collectively reflect the combination of epigenetic differences, integrated extrinsic signals, and molecular noise within individual neural crest cells (see [Figure 1a](#)) [8]. Such

Figure 1



The diversity of processes and features accounting for microheterogeneity in cell populations. (a) Subtypes of spatial, intrinsic, extrinsically-driven and temporal microheterogeneity hierarchically affecting the fate selection and phenotypic plasticity. Note that different subtypes of microheterogeneity may exist at distinct hierarchical levels and are not mutually exclusive. For instance, cell neighborhoods can significantly influence microheterogeneity through extrinsic signaling. Furthermore, cell fate bias and the competition between fate-related programs are shaped by multiple upstream mechanisms, including extrinsic signals, epigenetic memory, oscillatory dynamics, and molecular noise. Grey arrows illustrate hierarchical influences, while dotted arrows depict the interactions and mutual influences among the various types of microheterogeneity. (b) The explanatory scheme for the 3-step fate

microheterogeneity of biasing gene expression programs is the result of a simultaneous co-activation and then antagonistic repulsion of competing gene regulatory modules before a cell will commit to one of the available fates (also known as the 3-step mechanism [8], see Figure 1b and c). This process varies in each individual progenitor cell defining the final proportions of resulting cell types at a cell population scale, which is a key for mature tissue functional state and integrity. Via such mechanisms, the microheterogeneity of progenitors plays a crucial role in ensuring the generation of correct proportions of diverse cell types during neural development, particularly among the neural crest derivatives. In line with this, Figure 1C outlines different ways for a progenitor cell population to accumulate microheterogeneity while approaching a point of committing to a specific fate.

The comparison between probabilistic collective cell behaviors driven by plastic microheterogeneity and hard-wired fate choices brings into focus the different evolutionary advantages, flexibility, and robustness these strategies provide for developmental systems like the neural crest. In a probabilistic system where fate selection emerges from subtle, intrinsic biases and collective behaviors, cells can respond flexibly to environmental cues and adapt to changing conditions. This allows the system to dynamically adjust the proportions of different cell types based on the context, such as local tissue functions, signaling gradients, and extracellular environments. For example, microheterogeneity in neural crest cells ensures that a mixture of neuronal, glial, and mesenchymal cells can be produced in proportions that are adapted to the specific functions of the developing tissue [9,10]. In principle, a hard-wired system, where each cell acquires a fate based on on-off dynamics of a few powerful confounding mechanisms, offers far less adaptability. Such a system would be more rigid, and any perturbation - such as an injury, change in tissue environment, or genetic mutation - could disrupt the balance of cell types, leading to developmental defects or improper tissue function.

On the other hand, in some cases, microheterogeneity of cell states might be an inevitable outcome due to the numerous stochastic (random) factors that cells encounter during their development. This microheterogeneity might not necessarily confer a selective advantage; instead, it could be considered evolutionarily neutral or simply tolerated. The presence of this variability doesn't disrupt the overall system because cells have evolved regulatory mechanisms that maintain robust and orderly development despite such variations. While such microheterogeneity might be neutral, it

could also provide a buffer against environmental fluctuations or damage, offering some flexibility in the developmental programs.

Microheterogeneity due to individual cell history accumulated in epigenetic modifications

The most widely acknowledged microheterogeneity arises from individual cell histories, particularly through persisting epigenetic modifications. This type of microheterogeneity likely plays a crucial role in shaping the diversity of cell behaviors and fates within a population [11,12]. While cells may be genetically identical, the epigenetic landscape comprising DNA methylation, histone modifications, and chromatin structure varies significantly between individual cells based on their past experiences, exposures, and lineage histories [13,14]. These epigenetic changes can be stably inherited through cell divisions or be transiently altered in response to environmental cues, leading to functional differences within seemingly similar cells. Many examples of such epigenetic memory are known to play a role in the development of the CNS and neural crest lineage [9,15–18].

A key mechanistic example includes epigenetic priming mediated by modifications like histone methylation and acetylation, and the activity of Polycomb and Trithorax complexes, which altogether refer to the provisional establishment of epigenetic marks in neuroprogenitor cells, preparing the genome for future responses to signaling cues. These primed states do not immediately result in gene expression changes but set the stage for morphogen-induced differentiation into neural lineages [19]. Another key example is represented by bivalent regulatory chromatin domains, marked by both activating (H3K4me3) and repressive (H3K27me3) marks, which maintain biasing and cell differentiation genes in a "poised" state, allowing rapid activation upon receiving proper differentiation cues [20,21]. Such situations with epigenetic priming and bivalent enhancers might be differentially distributed in a population of neuroprogenitors and neural crest lineage cells, ensuring a dissimilar response to stimuli. For instance, the differential distribution of epigenetic marks in individual cells might result from stochastic differences in exposure to external signals throughout the cell lineage history.

Microheterogeneity of responses due to cell cycle and circadian oscillations

Depending on where a cell is in its cell cycle or circadian phase, it may have a temporary bias toward responding to certain cues. For example, a cell in a specific phase of

selection mechanism discovered in the neural crest cells [8]. (c) The progression of future fate-related microheterogeneity in cells before the commitment to a specific fate, according to the paradigm of the 3-step mechanism [8].

the cycle might be more sensitive to differentiation signals due to heightened expression of certain cell cycle-regulated genes. Although these phases are transient, they could combine with other signals to direct cell fate in a probabilistic manner [22,23]. Alternatively, during specific phases, cells might be insensitive to external influences, for instance, during the phase of compacted chromatin, when cells cannot effectively mount the transcriptional response. The degree of desynchronization of a cell cycle in a specific cell population might decide the proportion of cells responding or not responding to an extrinsic signal at a given time [22].

Furthermore, in zebrafish, neural crest cells migrate in organized streams, with leader cells undergoing faster cell cycle progression than follower cells [24]. These leader cells clonally give rise to sympathetic chain ganglia, while follower sister cells tend to differentiate into Schwann cells [25]. Such clonal competence might result from oscillating sensitivity to extrinsic signals during migration and requires further studies to check this hypothesis.

Also, the coordination between the cell cycle and fate decisions has been observed in the CNS radial glial cells, which must coordinate progenitor state amplification with neuronal differentiation to generate the multilayered cerebral cortex. Lineage commitment in this system depends on oscillating cyclins, with overexpression of CYCLIN-B1/B2 resulting in more upper-layer neurons at the expense of deep-layer neurons. Overexpression at later developmental stages produces fewer upper-layer neurons and more astrocytes, showing that developmental timing and cell cycle genes are interlinked [26].

In another example, oscillatory gene expression plays a crucial role in introducing heterogeneity and establishing distinct patterns in the previously uniform proliferating posterior embryonic tissue, such as neuromesodermal progenitors. This oscillatory process differentiates the dividing cells into somitic (future segments) and intersomitic (non-segmented) regions, which is essential for the proper formation of somites and the orderly elongation of the body axis during development [27–29]. In another example, the 2-h oscillations of HES1 have been shown to be crucial for the transition from neuronal progenitors to differentiated neurons and may play a significant role in neural cell fate decisions in a context of cell cycle length [30].

In a conceptually similar way, the circadian rhythms tightly regulate the expression of many genes involved in cell cycle control, DNA repair, and metabolism. In neural progenitors and adult neural stem cells, circadian clocks might cause oscillations in the expression levels of key neurogenic or gliogenic transcription factors,

growth factors, and signaling molecules [31–33]. Together with cell cycle oscillations, circadian cycles introduce a concept of a temporal microheterogeneity that could influence cell fate decisions, where cells in different phases respond differently to differentiation signals.

Microheterogeneity as the result of molecular noise

The heterogeneity of the cellular states approaching a cell fate decisions can be influenced by stochastic cell-intrinsic processes resulting in variation in the amount of proteins responsible for transcriptional activation, transport of mRNA, translation, mRNA, and protein degradation [34], collectively referring to as molecular noise, where transcriptional noise is a significant contributor [35]. Transcriptional noise is unavoidable and has been perceived as something that needs to be buffered or filtered to achieve a specific cellular phenotype [35]. However, recently transcriptional noise was suggested to have the potential to drive cell fate decisions by distorting the epigenetic landscape [36,37]. Also, noise can occur in signaling networks that do not involve transcription, such as those based on phosphorylation or changes in protein localization. Additionally, variability in the number of proteins distributed between daughter cells after division can also contribute to noise.

Molecular noise in general can influence cell fate decisions by flattening the landscape between attractors and allowing the cells to switch between the states. Of note, it's important to distinguish between variation (differences that exist) and chance (randomness). For example, when cells divide, even if they start with identical conditions (like flipping a coin), random fluctuations (chance) in protein levels or transcriptional noise can lead to different outcomes in each daughter cell, resulting in variation. This implies that variation can naturally arise due to random processes, even without initial differences or active regulatory mechanisms. Overall, noise isn't just a result of pre-existing variation but can increase due to the inherent randomness in biological processes [38–41].

This somehow resonates with how the traditional models of cell fate specification depicted differentiation as a series of discrete, stepwise transitions where cells move from one clearly defined state to another, whereas the more recent ATAC-seq single cell data and even the single cell transcriptomics reflect this process in a much more continual and “noisy” way [42]. This could mean that the cellular environment and stochastic processes (random fluctuations) play a larger role in fate determination than previously appreciated. The traditional clean-cut transitions are replaced by a more dynamic and probabilistic processes.

Microheterogeneity as the result of extrinsic signals

Microheterogeneity in a population of neural progenitor can arise as a direct result of extrinsic signals that originate from the cellular environment rather than from within the cells themselves. Even the slight variations in an extracellular matrix surrounding cells can affect the access to the signal and result in a differential response within a cell population [43,44]. This aligns with observations where the single-cell RNA sequencing, for instance, revealed specific gene modules in the chick neural plate border, where genes, previously considered specific markers of this region, exhibit dynamic expression [45]. This makes the area of the neural plate border not a defined line, but rather a contested territory, where cells express competing programs to decide their fate, and their identities are not yet fixed [46]. Such instability might be acting in many instances related to spatial patterning and fate selection. It might be, for instance, at work during the final sorting of the cell states between sister fates, for example, during establishing the ratios between neural crest-derived chromaffin cells, Schwann cells and sympathoblasts in the developing human adrenal glands. Potentially, in these adrenals, the extrinsic signal-driven microheterogeneity in progenitors arises due to mosaic activation of Htr3a/b ion channel receptor to serotonin, which probabilistically controls the proportions of generated terminal fates [47,48].

Furthermore, when modeled as a network, extrinsic cell signaling shows context-dependent information processing, enabling individual cells to respond uniquely to different stimuli depending on other signals and circumstances. This allows cells to make specific, context-dependent decisions, which is crucial for the precise regulation of cellular choices in multicellular organisms [49].

Microheterogeneity in self-organization and during signal-dependent patterning

Probabilistic and fluctuating expression of permissible gene modules in individual progenitors might, in principle, initiate the patterning of several domains of specifically-biased cells. This is sometimes observed in *in vivo* and *ex vivo* examples [50], including during abnormal and normal development of a floor plate and a roof plate in the neural tube [45]. Indeed, correct neural tube formation requires only one domain of a floor plate and of a roof plate [51]. The *in vitro* studies of this process uncovered the roles of domain competition [49], neighborhood-dependent mechanisms [52], and enhanced mutual inhibition [53], all associated with increasing microheterogeneity, gradual segregation, altogether ensuring the survival of the largest cell cluster [49,52]. Of note, such self-organizing mechanisms observed *in vitro* in controlled conditions are working in

conjunction with stochasticity and resulting microheterogeneity (operating within the context of broader regulatory mechanisms), as they seem to exist independently of the key extrinsic patterning signal - SHH from notochord - and can explain floor plate formation independently of notochord, for example, in the most anterior part of the embryo [49].

However, in the regions where SHH is secreted from the notochord, SHH acts as a buffering signal to minimize the microheterogeneity and stochasticity in the neuroprogenitor cells, which is key for refining the floor plate identity in a robust way [52].

Therefore, there are examples showing the ability of progenitor cells to break the symmetry and self-initiate the pattern formation even in the absence of the instructive action of the key morphogenetic signals, due to the probabilistic nature of expression of the permissive genes operating in the context of ready-to-go patterning and signaling mechanisms. The outcome of such self-organization based on microheterogeneity and assortative sorting may not be robust, because several abnormal cell groups of equal strength might emerge. In the absence of refining signals, the tiny differences in receptor expression, extracellular matrix properties, or signaling protein levels among progenitors might lead to differential sensitivity to external cues, amplifying those small initial differences into divergent fate outcomes.

Finally, microheterogeneity of any kind is not entirely autonomous; it relies on various regulatory mechanisms that interpret or manage it effectively. Each instance of microheterogeneity is associated with a set of molecular mechanisms that can either utilize or buffer it in a meaningful way. For example, in models where competing biasing programs influence fate selection, microheterogeneity operates within a relatively narrow range, tethered to the Waddingtonian landscape. This microheterogeneity works in conjunction with other fate selection mechanisms, enhancing the overall process. Together, these elements drive or increase the likelihood of specific fate decisions.

Phenotypic plasticity contributes to microheterogeneity

Phenotypic plasticity is the ability of cells to adopt different functional states in response to intrinsic or extrinsic cues. It is a key factor in generating dynamic microheterogeneity within populations of embryonic neuroprogenitors or postnatal tumor cells. This plasticity allows individual cells to transition between various phenotypic states, which leads to functional diversity even within a single cell type. Such dynamic behavior might contribute significantly to the complexity of cell fate decisions during development and cancer progression [54]. For instance, as neuroprogenitors progress

through the developmental process, their phenotypic plasticity allows them to respond to different sets of signals depending on the timing of their differentiation. This results in dynamic changes in their fate bias [55–57]. In tumors, particularly cancers such as glioblastoma or neuroblastoma, phenotypic plasticity plays a central role in creating dynamic microheterogeneity that drives tumor progression, metastasis, and resistance to treatment [58]. Tumor cells can transition between states of stemness, differentiation, and drug resistance, leading to fluctuating subpopulations of cells with different characteristics.

Microheterogeneity and combinatorial action of weak biasing signals of different nature

The concept of microheterogeneity joined with the combinatorial action of weak biasing signals offers a fascinating avenue for understanding how cell fate decisions are made. These signals - though individually weak or subtle - potentially could collectively create a robust bias driving cell fate outcomes within a progenitor population. By considering microheterogeneity across multiple axes, such as epigenetic differences, extrinsic signaling, molecular noise, and cell cycle/circadian rhythms, we can form a complex landscape of fate determination that is highly sensitive to initial conditions and contextual inputs.

Here we suggest a new perspective: imagine a multidimensional landscape where each axis represents a different weak biasing signal (epigenetic marks, extrinsic cues, transcriptional noise, etc.). Cells move through this landscape based on their internal states and external inputs. Each signal exerts a small force, gradually pushing cells toward different regions of the landscape, where specific fate outcomes are more likely. Even tiny initial differences between cells (due to microheterogeneity) will be amplified as cells integrate multiple signals over time. This creates a sensitive system where cells starting from very similar states can end up with different fates depending on the unique combination of weak signals they receive and compute. In this setting, weak signals might work best when combined with intrinsic noise (e.g., transcriptional noise), a phenomenon known as stochastic resonance [59]. In this scenario, noise does not disrupt the system but enhances it by helping weak signals overcome thresholds that would otherwise prevent the cell from committing to a fate.

Of note, in this new perspective we focus on the decision areas within grander Waddington's landscape, rather than on the structure of the landscape itself. The positioning of attractors and paths between attractors in this landscape is largely influenced by regulatory

mechanisms acting on the microheterogeneous cell population, an important point to consider. Therefore, when we discuss the new perspective on decision-making in a context of such landscape, we always operate within the limited fate decision area, instead of, for instance, the sharp decision point or global landscape spanning multiple fate decisions.

Conclusion: where the miniscule differences decide on big outcomes

Rather than distinct paths, cell fate decisions resemble a “cloud” of possible outcomes. This metaphor implies that there is a distribution of cell states, for which some future outcomes are more likely but not guaranteed, with probabilities shaped by subtle cellular variations. This is consistent with single cell RNA sequencing of neural crest progenitor population, which revealed differentially emerging biases in the expression of key regulatory genes that correlate with eventual fate decisions. The future perspectives shall focus on mathematical modeling and experimental perturbation addressing small shifts in expression of fate-biasing genes, which might reveal threshold-dependent switches in fate, with cells near the threshold showing stochastic behavior. The other future efforts are expected to focus on time-lapse imaging of single cells showing reporter activity reflecting fluctuating patterns of regulatory proteins prior to commitment of a cell to a specific fate, with the timing of fluctuations correlating with specific outcomes. The genetic recorders of incorporated signals acting on cells during fate selection based on CRISPR scarring or base editing might become another promising direction leading to better understanding of microheterogeneity-related agents and their combinatorial actions in cell fate decisions in developing nervous system.

Declaration of Competing interest

Authors declare no conflict of interest.

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Data availability

No data was used for the research described in the article.

References

Papers of particular interest, published within the period of review, have been highlighted as:

* of special interest

** of outstanding interest

1. Bronner ME, LeDouarin NM: **Evolution and development of the neural crest: an overview.** *Dev Biol* 2012, **366**:2–9.
2. Koontz A, Urrutia HA, Bronner ME: **Making a head: neural crest and ectodermal placodes in cranial sensory development.** *Semin Cell Dev Biol* 2023, **138**:15–27.
3. Streit A: **Early development of the cranial sensory nervous system: from a common field to individual placodes.** *Dev Biol* 2004, **276**:1–15.
4. Waddington CH: *The strategy of the genes.* Routledge; 2014.
5. Shimojo H, Isomura A, Ohtsuka T, Kori H, Miyachi H, Kageyama R: **Oscillatory control of Delta-like1 in cell interactions regulates dynamic gene expression and tissue morphogenesis.** *Genes Dev* 2016, **30**:102–116.
6. Azhar Y el, Schulthess P, Oostrom van MJ, Meijer WHM, Thomas WM, Bauer M, Sonnen KF: **Unravelling differential Hes1 dynamics during axis elongation of mouse embryos through single-cell tracking.** 2024, <https://doi.org/10.1101/2024.03.30.586691>.
7. Balázsi G, van Oudenaarden A, Collins JJ: **Cellular decision making and biological noise: from microbes to mammals.** *Cell* 2011, **144**:910–925.
8. Soldatov R, Kaucka M, Kastriti ME, Petersen J, Chontorotzea T, Englmaier L, Akkuratova N, Yang Y, Häring M, Dyachuk V, *et al.*: **Spatiotemporal structure of cell fate decisions in murine neural crest.** *Science* 2019, **364**, eaas9536.
9. Erickson AG, Kameneva P, Adameyko I: **The transcriptional portraits of the neural crest at the individual cell level.** *Semin Cell Dev Biol* 2023, **138**:68–80.
- The article discusses the role of single cell multi-omics in revealing the micro-heterogeneity and gene regulatory mechanisms that govern neural crest cell fate decisions. It focuses on novel experimental approaches to fully understand the connection between extrinsic signals and intracellular biases.
10. Soto J, Ding X, Wang A, Li S: **Neural crest-like stem cells for tissue regeneration.** *Stem Cells Transl Med* 2021, **10**:681–693.
11. Huang S: **Non-genetic heterogeneity of cells in development: more than just noise.** *Development* 2009, **136**:3853–3862.
12. Carter B, Zhao K: **The epigenetic basis of cellular heterogeneity.** *Nat Rev Genet* 2021, **22**:235–250.
- The article discusses how single-cell sequencing and epigenomic profiling revealed heterogeneity in gene expression and chromatin organization in the morphologically similar cells, and emphasizes the need for improved single-cell techniques to explore the epigenetic basis of cellular heterogeneity for better understanding of cell fate decisions.
13. Sadakierska-Chudy A, Kostrzewa RM, Filip M: **A comprehensive view of the epigenetic landscape Part I: DNA methylation, passive and active DNA demethylation pathways and histone variants.** *Neurotox Res* 2015, **27**:84.
14. Pujadas E, Feinberg A: **Regulated noise in the epigenetic landscape of development and disease.** *Cell* 2012, **148**:1123.
15. Hu N, Strobl-Mazzulla PH, Bronner ME: **Epigenetic regulation in neural crest development.** *Dev Biol* 2014, **396**:159–168.
16. Lossi L, Castagna C, Merighi A: **An overview of the epigenetic modifications in the brain under normal and pathological conditions.** *Int J Mol Sci* 2024, **25**:3881.
17. Delaney SP, Julian LM, Stanford WL: **The neural crest lineage as a driver of disease heterogeneity in Tuberous Sclerosis Complex and Lymphangioleiomyomatosis.** *Front Cell Dev Biol* 2014, **2**.
18. Kabbe M, Agirre E, Carlström KE, Pohl FB, Ruffin N, Bruggen van D, Meijer M, Seeker LA, Bestard-Cuche N, Lederer AR, *et al.*: **Single-nuclei histone modification profiling of the adult human central nervous system unveils epigenetic memory of developmental programs.** 2024, <https://doi.org/10.1101/2024.04.15.589512>.
19. Ernst C, Jefri M: **Epigenetic priming in neurodevelopmental disorders.** *Trends Mol Med* 2021, **27**:1106–1114.
20. Akkers RC, van Heeringen SJ, Jacobi UG, Janssen-Megens EM, François K-J, Stunnenberg HG, Veenstra GJC: **A hierarchy of H3K4me3 and H3K27me3 acquisition in spatial gene regulation in Xenopus embryos.** *Dev Cell* 2009, **17**:425–434.
21. Kumar D, Cinghu S, Oldfield AJ, Yang P, Jothi R: **Decoding the function of bivalent chromatin in development and cancer.** *Genome Res* 2021, **31**:2170–2184.
- Bivalent chromatin, marked by the simultaneous presence of H3K4me3 and H3K27me3, protects promoters from DNA methylation instead of marking genes for rapid activation. Loss, of H3K4me3, may lead to hypermethylation and irreversible silencing in diseases such as cancer.
22. Snijder B, Sacher R, Rämö P, Damm E-M, Liberali P, Pelkmans L: **Population context determines cell-to-cell variability in endocytosis and virus infection.** *Nature* 2009, **461**:520–523.
23. Battich N, Stoeger T, Pelkmans L: **Control of transcript variability in single mammalian cells.** *Cell* 2015, **163**:1596–1610.
24. Alhashem Z, Feldner-Busztin D, Revell C, Portillo MAG, Camargo-Sosa K, Richardson J, Rocha M, Gauert A, Corbeaux T, Milanetto M, *et al.*: **Notch controls the cell cycle to define leader versus follower identities during collective cell migration.** *Elife* 2022, **11**.
25. Alhashem Z, Camargo-Sosa K, Kelsh RN, Linker C: **Trunk neural crest migratory position and asymmetric division predict terminal differentiation.** *Front Cell Dev Biol* 2022, **10**.
- Research shows that migratory identity strongly correlates with terminal differentiation. Leader cells, which guide migration, divide asymmetrically, with one daughter cell retaining the leader identity and forming sympathetic chain ganglia, while the other adopts a follower identity, differentiating into Schwann cells. These processes are regulated by transcription factors like phox2bb and environmental signals.
26. Hagey DW, Topcic D, Kee N, Reynaud F, Bergsland M, Perlmann T, Muhr J: **CYCLIN-B1/2 and -D1 act in opposition to coordinate cortical progenitor self-renewal and lineage commitment.** *Nat Commun* 2020, **11**:2898.
27. Sambasivan R, Steventon B: **Neuromesodermal progenitors: a basis for robust axial patterning in development and evolution.** *Front Cell Dev Biol* 2021, **8**, 607516.
28. Wanglar C, Takahashi J, Yabe T, Takada S: **Tbx protein level critical for clock-mediated somite positioning is regulated through interaction between Tbx and Ripply.** *PLoS One* 2014, **9**, e107928.
29. William DA, Saitta B, Gibson JD, Traas J, Markov V, Gonzalez DM, Sewell W, Anderson DM, Pratt SC, Rappaport EF, *et al.*: **Identification of oscillatory genes in somitogenesis from functional genomic analysis of a human mesenchymal stem cell model.** *Dev Biol* 2007, **305**:172–186.
30. Kobayashi T, Kageyama R: **Expression dynamics and functions of Hes factors in development and diseases.** *Curr Top Dev Biol* 2014, **110**:263–283.
31. Bery A, Bagchi U, Bergen AA, Felder-Schmittbuhl M-P: **Circadian clocks, retinogenesis and ocular health in vertebrates: new molecular insights.** *Dev Biol* 2022, **484**:40–56.
- The article discusses the role of circadian rhythms in retinal development and their influence on the timing of precursor cell proliferation and differentiation, and suggests that understanding these mechanisms could improve therapies for retinal diseases, including time-based drug administration.
32. Arafa K, Emara M: **Insights about circadian clock and molecular pathogenesis in gliomas.** *Front Oncol* 2020, **10**:199.
33. Draijer S, Chaves I, Hoekman MFM: **The circadian clock in adult neural stem cell maintenance.** *Prog Neurobiol* 2019, **173**: 41–53.
34. Urban EA, Johnston RJ: **Buffering and amplifying transcriptional noise during cell fate specification.** *Front Genet* 2018, **9**.

35. Arias AM, Hayward P: **Filtering transcriptional noise during development: concepts and mechanisms.** *Nat Rev Genet* 2006, **7**:34–44.
36. Xue G, Zhang X, Li W, Zhang L, Zhang Z, Zhou X, Zhang D, Zhang L, Li Z: **A logic-incorporated gene regulatory network deciphers principles in cell fate decisions.** *Elife* 2023, **12**.
 The article presents how noise-driven and signal-driven forces uniquely influence fate bias, reprogramming trajectories, and differentiation stages, providing a basis for understanding cell fate decisions in hematopoiesis, embryogenesis, and trans-differentiation through noise patterns and regulatory logic.
37. Coomer MA, Ham L, Stumpf MPH: **Noise distorts the epigenetic landscape and shapes cell-fate decisions.** *Cell Syst* 2022, **13**: 83–102.e6.
38. Elowitz MB, Levine AJ, Siggia ED, Swain PS: **Stochastic gene expression in a single cell.** *Science* 2002, **297**:1183–1186.
39. Paulsson J: **Summing up the noise in gene networks.** *Nature* 2004, **427**:415–418.
40. Tendler A, Zimmer A, Mayo A, Alon U: **Noise-precision tradeoff in predicting combinations of mutations and drugs.** *PLoS Comput Biol* 2019, **15**, e1006956.
41. Sengupta S, Das S, Crespo AC, Cornel AM, Patel AG, Mahadevan NR, Campisi M, Ali AK, Sharma B, Rowe JH, et al.: **Mesenchymal and adrenergic cell lineage states in neuroblastoma possess distinct immunogenic phenotypes.** *Nat Can (Ott)* 2022, **3**:1228–1246.
42. Sagar Grün D: **Deciphering cell fate decision by integrated single-cell sequencing analysis.** *Annu Rev Biomed Data Sci* 2020, **3**:1.
43. Rozario T, DeSimone DW: **The extracellular matrix in development and morphogenesis: a dynamic view.** *Dev Biol* 2010, **341**:126–140.
44. Gattazzo F, Urciuolo A, Bonaldo P: **Extracellular matrix: a dynamic microenvironment for stem cell niche.** *Biochim Biophys Acta BBA - Gen Subj* 2014, **1840**:2506–2519.
45. Thiery AP, Buzzi AL, Hamrud E, Cheshire C, Luscombe NM, Briscoe J, Streit A: **scRNA-sequencing in chick suggests a probabilistic model for cell fate allocation at the neural plate border.** *Elife* 2023, **12**, e82717.
 Neural plate border (NPB) in chick embryos contains a heterogeneous population of progenitor cells expressing competing transcriptional programs for neural, neural crest, and placodal fates, with fate decisions occurring progressively as these "Border Located Undecided Progenitors" (BLUPs) commit into specific lineages during neurulation.
46. Kotov A, Seal S, Alkobtawi M, Kappès V, Ruiz SM, Arbès H, Harland RM, Peshkin L, Monsoro-Burq AH: **A time-resolved single-cell roadmap of the logic driving anterior neural crest diversification from neural border to migration stages.** *Proc Natl Acad Sci* 2024, **121**, e2311685121.
47. Engel M, Smidt MP, van Hooft JA: **The serotonin 5-HT3 receptor: a novel neurodevelopmental target.** *Front Cell Neurosci* 2013, **7**:76.
48. Murthy S, Niquille M, Hurni N, Limoni G, Frazer S, Chameau P, van Hooft JA, Vitalis T, Dayer A: **Serotonin receptor 3A controls interneuron migration into the neocortex.** *Nat Commun* 2014, **5**:5524.
49. Krammer T, Stuart HT, Gromberg E, Ishihara K, Cislo D, Melchionda M, Becerril Perez F, Wang J, Costantini E, Lehr S, et al.: **Mouse neural tube organoids self-organize floorplate through BMP-mediated cluster competition.** *Dev Cell* 2024, **59**:1940–1953.e10.
50. Symmons O, Raj A: **What's luck got to do with it? Single cells, multiple fates, and biological non-determinism.** *Mol Cell* 2016, **62**:788–802.
51. Lehr S, Brückner DB, Greunz-Schindler M, Minchington T, Merrin J, Hannezo E, Kicheva A: **Self-organised pattern formation in the developing neural tube by a temporal relay of BMP signalling.** 2023, <https://doi.org/10.1101/2023.11.15.567070>.
52. Abdel Fattah AR, Grebenyuk S, de Rooij LPMH, Salmon I, Ranga A: **Neuroepithelial organoid patterning is mediated by a neighborhood watch mechanism.** *Cell Rep* 2023, **42**, 113334.
53. Rekler D, Kalcheim C: **Completion of neural crest cell production and emigration is regulated by retinoic-acid-dependent inhibition of BMP signaling.** *Elife* 2022, **11**, e72723.
54. Whiting FJH, Househam J, Baker A-M, Sottoriva A, Graham TA: **Phenotypic noise and plasticity in cancer evolution.** *Trends Cell Biol* 2023, <https://doi.org/10.1016/j.tcb.2023.10.002>.
55. Kohwi M, Doe CQ: **Temporal fate specification and neural progenitor competence during development.** *Nat Rev Neurosci* 2013, **14**:823–838.
 Neural progenitors can undergo transcriptional and chromatin changes to regulate temporal identity and competence during development, enabling the sequential specification of diverse CNS cell types.
56. Coquand L, Brunet Avalos C, Macé A-S, Farcy S, Di Cicco A, Lampic M, Wimmer R, Bessières B, Attie-Bitach T, Fraissier V, et al.: **A cell fate decision map reveals abundant direct neurogenesis bypassing intermediate progenitors in the human developing neocortex.** *Nat Cell Biol* 2024, **26**:698–709.
57. Di Bella DJ, Habibi E, Stickels RR, Scalia G, Brown J, Yadollahpour P, Yang SM, Abbate C, Biancalani T, Macosko EZ, et al.: **Molecular logic of cellular diversification in the mouse cerebral cortex.** *Nature* 2021, **595**:554–559.
58. Gupta PB, Pastushenko I, Skibinski A, Blanpain C, Kuperwasser C: **Phenotypic plasticity as a driver of cancer formation, progression and resistance to therapy.** *Cell Stem Cell* 2019, **24**:65–78.
 Authors explore the concept of phenotypic plasticity in adult somatic cells, particularly in the context of cancer initiation, progression, and therapy resistance, emphasizing the importance of cellular dedifferentiation.
59. McDonnell MD, Abbott D: **What is stochastic resonance? Definitions, misconceptions, debates, and its relevance to biology.** *PLoS Comput Biol* 2009, **5**, e1000348.